

October Night Sky Brings Hidden Jupiter and Meteor Showers

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October's longer nights give skywatchers several events to look for, NASA notes. Among them are a planetary disappearing act and two **meteor showers**.

Before sunrise on October 6, people in parts of eastern and central North America could see Jupiter slip behind the Moon's thin crescent for about an hour. Astronomers call this a lunar **occultation** when the Moon blocks a more distant object. The event was visible without a telescope, though binoculars helped. Timing varied by location; in some places it happened around 4 a.m. Eastern time.

The Draconid **meteor shower** peaked on the evening of October 8 into early October 9. Unlike many showers that are best after midnight, the **Draconids** can be watched soon after nightfall, around 9 p.m. Eastern. They appear to come from the constellation Draco, the dragon. The display is usually modest only a handful of meteors per hour but this year's dark skies near new Moon helped. The best views come away from city lights after eyes adjust to the dark.

Another shower, the **Orionids**, peaks around October 21. Those meteors are bits of debris from Halley's Comet. When Earth passes through the trail, the particles burn up as bright streaks. NASA estimates suburban viewers may see about 3 to 6 meteors per hour, with more under darker skies. The **Orionids** seem to radiate from Orion. A bright **waxing gibbous** Moon may hide fainter meteors until the Moon sets around 3 a.m. local time, when Orion is higher before sunrise.

No special equipment is required for these shows just clear skies, patience, and a safe dark place to look up.

Word Treasure

occultation -- When one sky object temporarily hides another from our view.

meteor shower -- A period when many shooting stars appear from one sky direction.

Draconids -- An October meteor shower that seems to come from the dragon constellation Draco.

Orionids -- A meteor shower linked to debris from Halley's Comet near Orion.

waxing gibbous -- A Moon phase that is more than half lit and still growing brighter.

Background you need

Meteor showers happen when Earth passes through trails of dusty debris left by comets. The Moon can hide planets or stars for a short time when its path lines up just right. Darker skies make faint meteors easier to see, while bright moonlight can wash them out. Many sky events can be enjoyed without telescopes if you have clear weather and a safe viewing spot.

Break it down

LEAD: October offers occultation and two meteor showers

EVENT 1: Jupiter lunar occultation on October 6

EVENT 2: Draconids peak October 8-9 after nightfall

EVENT 3: Orionids peak around October 21 from Halley debris

VIEWING TIPS: Dark skies; Moon may interfere with Orionids until ~3 a.m.

OPEN QUESTION: How many meteors each observer will actually catch

Why it matters

Seasonal sky events invite kids to practice careful looking and connect calendar dates with real astronomy. Knowing when moonlight helps or hurts viewing makes nights outdoors more successful.

Test what you remember

1. What happened to Jupiter on October 6?

- ☐ (A) It slipped behind the Moon for about an hour
- ☐ (B) It turned into a comet
- ☐ (C) It crashed into Earth
- ☐ (D) It exploded

2. When did the Draconids peak?

- ☐ (A) December 25
- ☐ (B) October 8 and 9
- ☐ (C) July 4
- ☐ (D) Only at noon

3. Where do Orionid meteors come from?

- ☐ (A) Pieces of the Moon
- ☐ (B) Airplane lights
- ☐ (C) Debris from Halley's Comet
- ☐ (D) Weather balloons

4. About how many Orionids might suburban viewers see per hour?

- ☐ (A) Zero always
- ☐ (B) One thousand
- ☐ (C) Exactly 100
- ☐ (D) About 3 to 6

5. Why can Draconids be easier for casual watchers?

- ☐ (A) They only appear at sunrise from space stations
- ☐ (B) They can be seen soon after nightfall
- ☐ (C) They require a spacesuit
- ☐ (D) They last all year at noon

6. What may hide fainter Orionids this year?

- ☐ (A) A solar eclipse all night
- ☐ (B) Heavy snow in every city
- ☐ (C) Airport closures
- ☐ (D) A bright waxing gibbous Moon

Answer key (try the quiz first!): 1: A 2: B 3: C 4: D 5: B 6: D

MY THOUGHTS (at least 20 words):